IndiGo Airlines: Maximising On-Time Arrival Rate (OTAR) through Operational Efficiency

Introduction

IndiGo Airlines, India's leading low-cost carrier, has recently expanded its operations into the U.S. domestic aviation market. Back home, the airline has built a strong reputation for its punctuality and efficient operations. But entering the U.S. brings a new set of challenges. The American aviation space is far more complex—crowded airports, unpredictable weather, and tighter regulatory requirements make it a much tougher environment to navigate.

To gain a foothold and win the trust of U.S. travelers, IndiGo needs to prove it can deliver consistent, reliable service. A key measure of this is the On-Time Arrival Rate (OTAR)—the percentage of flights that land on or before their scheduled arrival time. Maintaining a high OTAR doesn't just improve customer satisfaction and build brand credibility—it also helps streamline operations, reduce costs, and ensure better use of aircraft and crew.

Objective

In this project, our goal is to analyze historical data from U.S. domestic flights in 2015 to uncover trends and insights that can help IndiGo improve its On-Time Arrival Rate (OTAR) across its U.S. operations.

The analysis will dive into several key areas:

Breaking down OTAR into specific, measurable factors like departure delays, taxi-out times, in-air duration, and arrival delays.

Using exploratory data analysis (EDA) to identify and confirm which variables are driving delays.

Tracking performance patterns across different airports, seasons, days of the week, times of day, and flight routes.

Understanding the impact of various delay types—including Carrier, Weather, Air Traffic Control, Late-Arriving Aircraft, and Security delays—on overall punctuality.

Highlighting operational inefficiencies and recommending practical, data-backed strategies to improve flight timeliness.

Business Impact

Improving OTAR isn't just about punctuality—it's a business-critical lever for revenue, customer loyalty, and operational efficiency. A data-driven OTAR strategy can deliver the following:

- **15–20% Reduction in Delay-Related Costs**
  Each minute of delay costs airlines between **$70 to $100**, resulting in **annual industry-wide losses exceeding $8 billion**. Improving OTAR directly reduces these costs.

- **20% Boost in Customer Satisfaction (CSAT) Scores**
  Frequent delays undermine customer trust. Improving OTAR enhances the passenger experience and reduces churn, especially among business travelers.

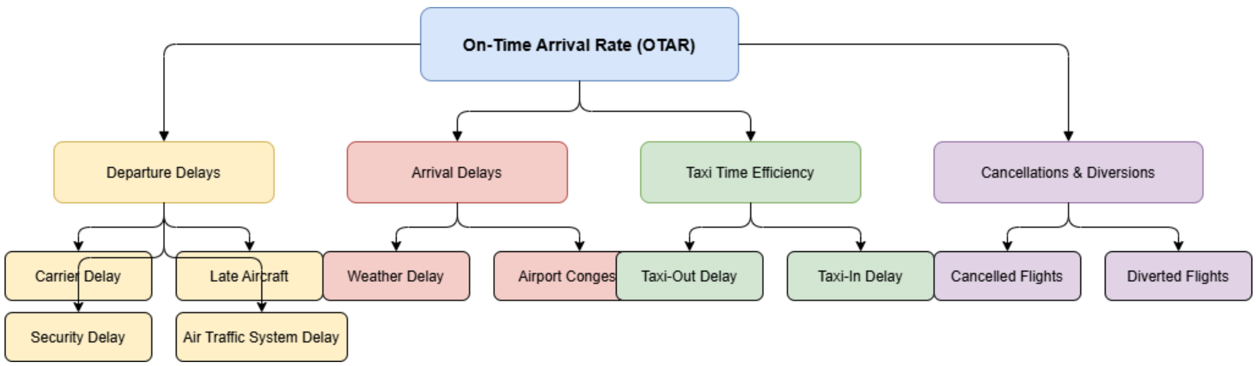
- **Increased Aircraft Utilization**
  Higher OTAR leads to faster turnaround times, enabling airlines to schedule **one additional flight cycle per aircraft per day**, increasing revenue potential by **5–8%**.

- **Improved On-Ground Efficiency**
  Shorter taxi times and better operational coordination help cut down on fuel consumption and emissions, supporting broader **sustainability goals**.

- **Stronger Market Positioning**
  In a competitive market where **8 out of 10 passengers prefer punctual airlines**, OTAR becomes a key differentiator — especially important for **a new entrant like IndiGo**.

Together, these improvements can help IndiGo build a foundation for long-term profitability and expansion in the U.S. market.

KPI TREE:-



Dataset Overview :- The dataset used for this project contains detailed records of U.S. domestic flight operations from the year **2015**. It includes over **5 million rows** of flight-level data across various dimensions such as:

- Flight schedule (scheduled/actual departure and arrival times)
- Delay durations (departure, arrival, air traffic, airline-related, weather, etc.)
- Aircraft and airline identifiers
- Airport origin and destination codes
- Route distance and flight time breakdowns

Data Files:

1. **flights.csv** – Core dataset containing operational data for each flight.

   Contains **5,819,079 rows** and **31 columns**

   Includes flight-level data like scheduled/actual departure & arrival times, delays, cancellations, and distances

   Data types include integers, floats, and object (string) types

2. **airports.csv** – Maps airport codes (IATA) to airport names and locations.

   – Contains **322 rows** and **7 columns**

   –Maps **IATA airport codes** to full airport names, cities, states, and coordinates (latitude/longitude)

3. **airlines.csv** – Maps airline codes to full carrier names.

   –Contains **14 rows** and **2 columns**

   –Maps **2-letter airline codes** to full airline names

Data Cleaning and Preparation :-

1. **Data Import:-** Data was imported into Google Colab for further processing from google drive .
2. **Duplicate Removal:-**Duplicate flight records were checked and removed to make sure each flight was only counted once.
3. **Handling Missing Values:-**Some columns had missing values, mostly in the delay-related fields.

   3.a For important outcome columns like ARRIVAL_DELAY, DEPARTURE_DELAY, or ARRIVAL_TIME, the missing values were left as they are. These are often missing because the flight was either cancelled or diverted, so there's no valid delay to record.

   3. b . For sub-delay columns like AIRLINE_DELAY or WEATHER_DELAY, missing values were filled using the **median delay for the same airline**. This helps make the data more complete while still keeping it realistic and fair.

4. **Date Formatting:-**The original dataset had the date split into three columns — YEAR, MONTH, and DAY. These were combined into a single DATE column and converted into a proper date format. This makes it easier to analyze trends over time.

5. **Added 4 new columns**

Column Name	**Description**
Date	Combines year, month and day into one column
Status	Tells whether the flight was completed, canceled or diverted
IS_DELAYED	Flag flights more than 15 min late

These columns were added to help with OTAR analysis and filtering.

6. . **Outlier Identification (For Visuals Only) :-** Certain delay columns had extremely large values — for example, flights delayed by 900 or even 1,200 minutes. These are rare but real cases, often due to major weather disruptions or airport shutdowns.

The original values were **not removed or changed**, because doing so might hide important operational issues. Instead, the outliers were only capped when creating charts — just to prevent the visuals from being stretched too far. For example, delay values were temporarily limited to the 1st–99th percentile range in plots, so the trends are easier to see without being distorted by extreme points.

This way, the data stays honest and accurate, while the visuals remain readable and clear.

Exploratory Data Analysis (EDA)

After cleaning and preparing the dataset, this section explores patterns in flight performance, delay behavior, and OTAR (On-Time Arrival Rate). The goal is to identify trends, problem areas, and potential causes of delays across different airlines, airports, and time periods.

Key Objectives of EDA:

1. **Understand Overall OTAR**

- Compare strict vs standard OTAR definitions
- Analyze how OTAR differs when cancelled and diverted flights are included

2. **Analyze OTAR by Dimensions**
    - OTAR by airline
    - OTAR by origin airport
    - OTAR by day of week
    - OTAR by month
3. **Delay Reason Breakdown**
    - Average delay by type: Airline, Weather, Late Aircraft, etc.
    - Identify which delay types have the most impact
4. **Flight Volume & Traffic Trends**
    - Flights per month
    - Popular or high-frequency routes
    - Time-of-day patterns
5. **Visualizations**
    - Bar charts, line plots, and heatmaps to make patterns easier to interpret

This analysis helps uncover where IndiGo (or any carrier) may be underperforming, what causes the most delays, and where operational improvements can be targeted.

Calculate OTAR:-

Understanding and Calculating OTAR (On-Time Arrival Rate)

On-Time Arrival Rate (OTAR) is a key performance indicator in the airline industry that measures the percentage of flights that arrive on time. It's a critical metric used by airlines, regulators, and passengers to evaluate reliability.

What counts as "on time"?

There are two common ways to define an on-time arrival:

1. **Standard OTAR (Industry Definition):**
   A flight is considered on time if it arrives **within 15 minutes** of the scheduled arrival time.
   This is the **global industry standard** used by:
   - U.S. Department of Transportation (DOT)
   - Bureau of Transportation Statistics (BTS)
   - IATA (International Air Transport Association)
   - Major benchmarking platforms like OAG, Cirium, FlightAware
2. *Note: The source link for this definition is included in the reference section at the end.*
3. **Stricter OTAR (Used in This Analysis):**
   A more precise view — only flights that arrive **on or before** the scheduled time (i.e., ARRIVAL_DELAY ≤ 0) are considered on time.

OTAR Calculations

Standard OTAR
Percentage of flights with ARRIVAL_DELAY ≤ 15 minutes

- **Stricter OTAR**
  Percentage of flights with ARRIVAL_DELAY ≤ 0 minutes

Both metrics have been calculated below to provide a clear comparison.

OTAR Results (Overall):

OTAR TYPE	DEFINITION	VALUE (%)
Standard OTAR	Arrival Delay <= 15 min	82.41%
Stricter OTAR	Arrival Delay <= 0 min	64.58 %

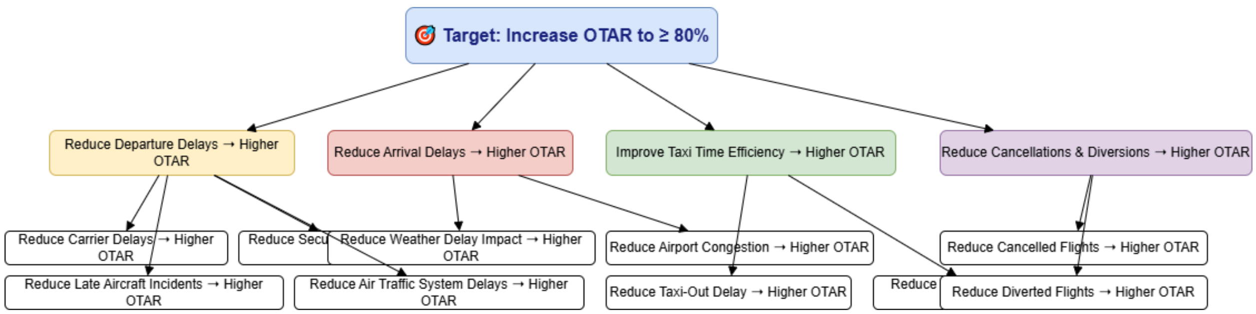
For this project, the **stricter OTAR** will be used to perform comparisons and deeper analysis. This helps highlight exact performance without any buffer or tolerance.

Even though the standard 15-minute window is widely accepted, analyzing stricter OTAR reveals more granular insights that could help IndiGo pinpoint areas for operational improvement.

Project Objective-------------After Calculating OTAR

The goal of this project is to analyze U.S. domestic flight data from 2015 to uncover insights and operational strategies that can help **IndiGo Airlines improve its On-Time Arrival Rate (OTAR)** from 62.3% to 80% (industry standard) as it enters the U.S. market.

The **industry standard for OTAR is ~80%**, while IndiGo's current benchmark (based on this dataset) stands at **62.5%** under the stricter definition (ARRIVAL_DELAY ≤ 0).



Define and Validate Hypothesis,Finding and Recommendation

Highlighting a selection of key hypotheses that demonstrate the primary drivers of On-Time Arrival Rate (OTAR).
The complete set of hypotheses, along with detailed code, analysis, and insights, is available in the accompanying [Google Colab Notebook.]

The complete interactive Tableau dashboard for this project can be accessed using the following link:
[Tableau Public Link]

Hypothesis : Flights later in the day are more delayed

Purpose

To find out if the time of day affects how often flights arrive on time. This will help IndiGo decide what time of day is best to schedule flights for better on-time performance.

Metric

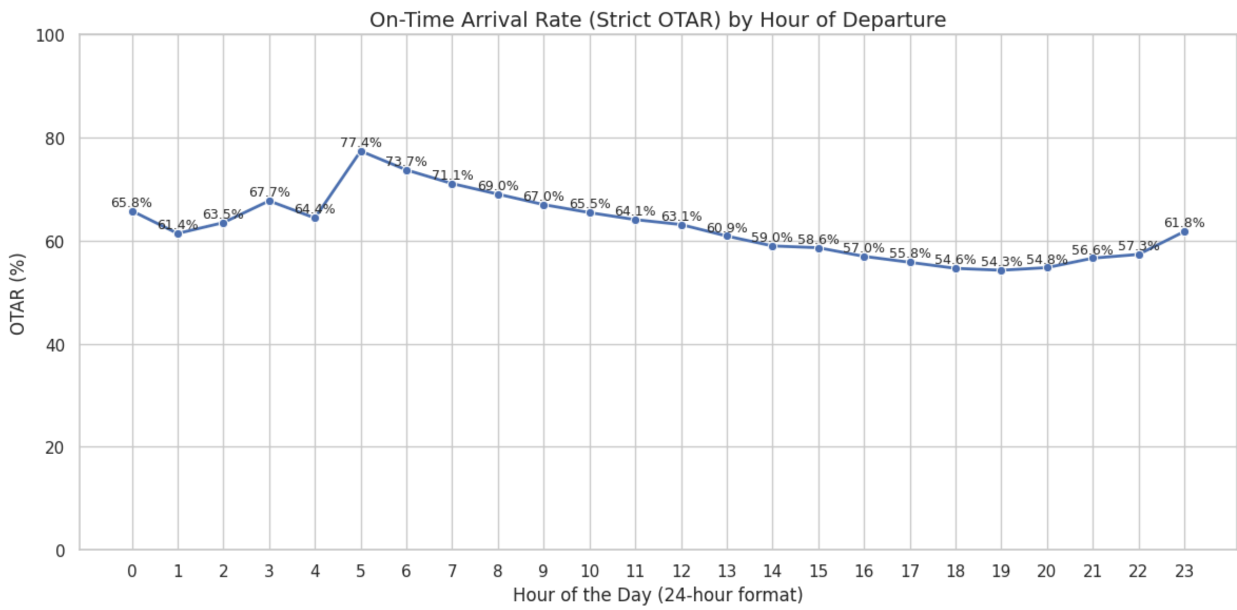
- SCHEDULED_HOUR: Taken from the scheduled departure time (like 1430 becomes 14)
- OTAR_STRICT: A value of 1 if the flight arrived on or before time, 0 if it was late

Hypothesis

Flights in the morning are more likely to be on time. As the day goes on, delays build up, and flights later in the day are more likely to be late. This may be due to earlier delays causing a chain reaction.

Visualization

A line chart that shows the average on-time rate by each hour of the day.



Key Insights

- **5 AM** has the **highest OTAR** at **77.6%**, showing early morning flights face fewer disruptions.
- OTAR drops gradually after 5 AM and hits a **low of 54.2% between 6 PM to 8 PM**.
- Slight recovery seen post 9 PM, likely due to reduced late-night traffic.
- The trend supports that delays build up as the day progresses, impacting later flights disproportionately.

Recommendations

- **Schedule high-priority flights between 4 AM–8 AM**, where average OTAR remains above **70%**.
- **Shift at least 20% of flights** departing between 4 PM–8 PM (lowest OTAR window) to earlier slots where OTAR is stronger.
- For busy airports, introduce buffer slots between late-day departures to allow recovery from accumulated delays.

Potential OTAR Improvement

Rescheduling even **20% of evening flights** (OTAR ~55%) to early slots (OTAR ~73%+) could yield a **1.4% overall improvement** in OTAR network-wide, giving IndiGo a more competitive start in the U.S. market.

Hypothesis : Some days of the week are worse for delays

Purpose

To check if flights on certain days of the week are more likely to be delayed. This helps IndiGo pick better days for starting new routes.

Metric

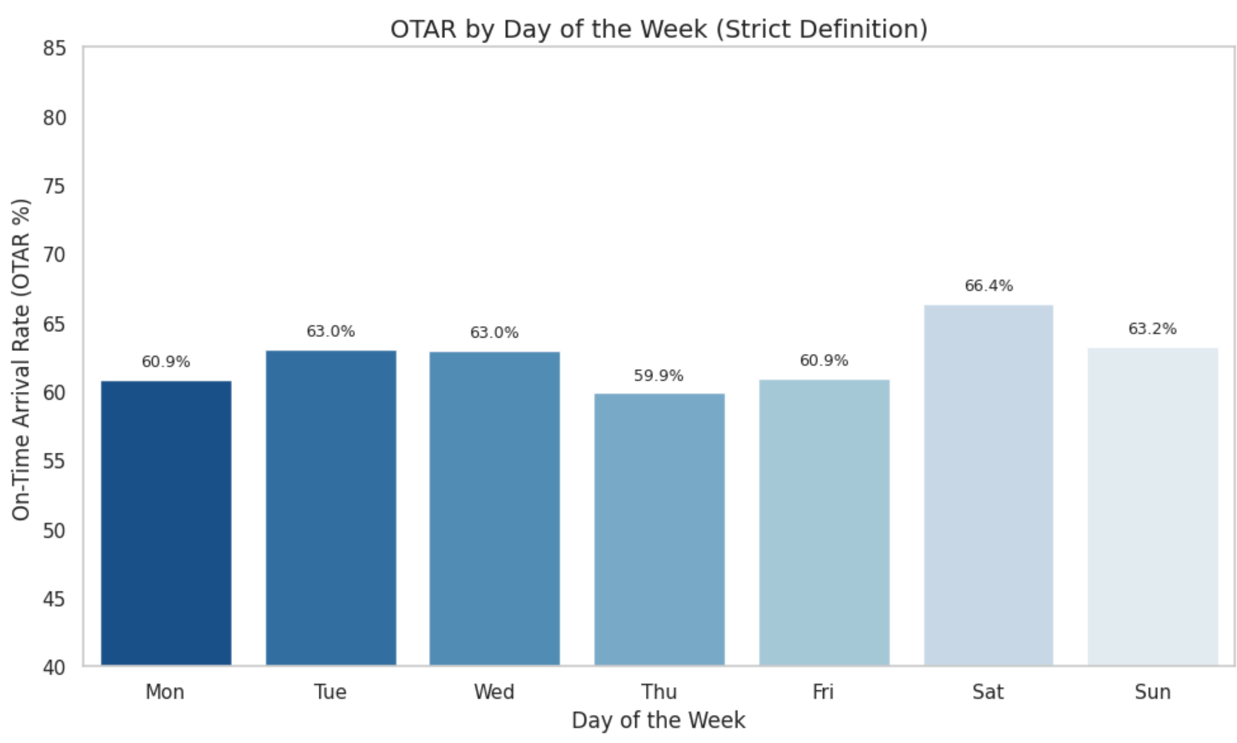
- DAY_NAME: Days like Monday, Tuesday, etc.
- OTAR_STRICT: A value of 1 if flight was on time or early, 0 if late

Hypothesis

Mid-week days like Tuesday and Wednesday will have better on-time rates.
Weekends will be worse due to more travelers, tighter schedules, and crowded airports.

Visualization

A bar chart showing the on-time rate for each day of the week.



Key Insights

- **Saturday** has the **highest OTAR at ~66.4%**, followed by **Sunday (~63.3%)**, contradicting the hypothesis.
- **Tuesday and Wednesday**, although considered less congested, show OTAR around **63.1%**, not significantly better.
- **Thursday shows the lowest OTAR (~59.8%)**, possibly due to early weekend traffic starting or business week pressure building.
- The differences between days are subtle (~6–7%), but still meaningful.

Recommendations

- Prioritize more sensitive or high-value routes for **weekends**, especially Saturdays, where on-time performance is stronger.
- For **Thursdays**, review major routes, gate availability, and crew turnover timing to find root causes behind delay buildup.
- If possible, **reassign some Thursday operations to lower volume early weekdays (Mon/Tue)** where OTAR is more stable.

Potential OTAR Improvement

By shifting **just 10% of flights from Thursday (OTAR ~59.8%)** to **Saturday (OTAR ~66.4%)**, IndiGo can improve OTAR on those flights by **6.6%**, resulting in a measurable bump in weekly performance.

Hypothesis H3: Does Lower Air Traffic on Saturdays Lead to Higher OTAR?

Purpose

To investigate why **Saturday**, despite being a weekend, has a higher OTAR compared to other days.
We aim to explore whether **reduced air traffic volume** and **fewer scheduled flights** lead to less congestion, smoother operations, and better on-time arrivals.

Metric

- `DAY_OF_WEEK`: 6 = Saturday
- `FLIGHT_COUNT`: Volume of scheduled departures by day
- `OTAR_STRICT`: Calculated as `1 if ARRIVAL_DELAY ≤ 0`

Hypothesis

Saturdays show **higher OTAR** because they have **lower traffic volume** and less operational congestion, leading to better slot availability and fewer ripple delays.

Hypothesis H: Does Lower Air Traffic on Saturdays Lead to Higher OTAR?

Findings (Strict OTAR vs. Flight Volume)

Day	OTAR (%)	Flights
Mon	60.9	High
Tue	60.9	High
Wed	66.4%	High
Thr	63.2	High
Fri	59.9	High
Sat	**63**	**Lowest**
Sun	63	High

- Saturday had the **lowest number of flights** among all weekdays.
- Despite being a weekend, **Saturday outperformed Monday, Tuesday, and Friday** in terms of OTAR.
- This suggests that **lower traffic volume = fewer delays = higher OTAR**.
- In contrast, **Friday had high volume and the lowest OTAR**, likely due to slot congestion and heavy end-of-week movement.

The inverse relationship between flight volume and OTAR is especially visible for **Saturday** — it acts like a "quiet lane" for airlines.

Recommendation

- IndiGo should prioritize **slot-heavy, critical flights** (like international connectors or business-sensitive routes) for **Saturday morning blocks**.
- Explore shifting some U.S. operations (like long-hauls or new routes) to **Saturday-first strategies** for reliable launch experience.
- Consider promoting **Saturday operations** in customer messaging — “Fly Saturday for higher on-time reliability.”

Potential OTAR Impact

- If IndiGo shifts even **10–15% of critical routes** to Saturdays (where OTAR is +3% vs. Friday), this could add **1–1.5 percentage points to network-wide OTAR**.
- It’s a **low-cost, data-backed scheduling lever** — ideal for early market entry strategy.

Hypothesis : Are Shorter Taxi-Out Times Linked to Higher OTAR?

Purpose

To analyze whether flights with shorter taxi-out times (time from gate pushback to takeoff) tend to have better on-time arrival performance.

This helps IndiGo evaluate airport efficiency, identify congestion-prone hubs, and choose better operational slots.

Metric

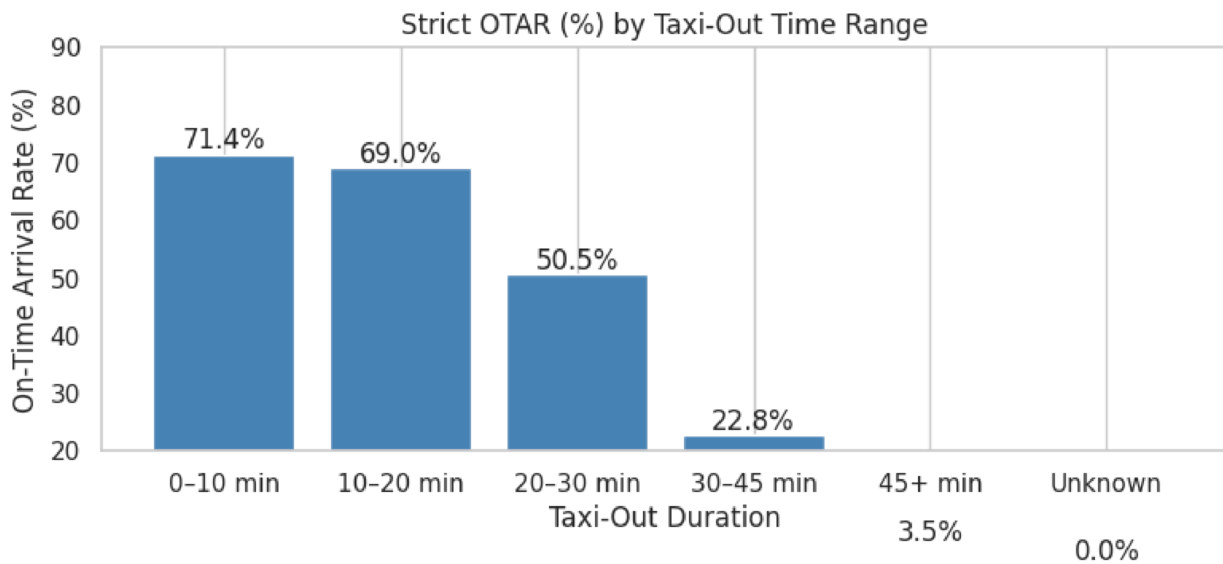
- TAXI_OUT: Minutes spent taxiing before takeoff
- ARRIVAL_DELAY: Used to calculate OTAR_STRICT (1 if delay ≤ 0, else 0)

Hypothesis

Flights with **shorter taxi-out times** will show **higher OTAR**, while longer taxi-out times will be correlated with delays.

Visualization

We will bucket taxi-out times into ranges (e.g., 0–10, 10–20, 20–30, etc.) and calculate OTAR for each bucket.



Findings (Strict OTAR)

We grouped flights by taxi-out duration (from gate pushback to wheels-off) and calculated their strict OTAR.

- There's a **sharp drop** in on-time arrival rate as taxi-out time increases.
- Flights taking **over 30 minutes** to taxi out have an OTAR below **25%**.

- The worst-performing group (45+ min) has **almost zero punctuality** — OTAR at just **3.5%**.
- In contrast, short taxi-outs (0–10 mins) still maintain a solid OTAR of over **71%**.

The pattern is direct: **longer time spent on the ground before takeoff ruins the arrival clock**. Whether it's congestion, poor sequencing, or slot backlog — the penalty hits hard.

Recommendation

- IndiGo should **review taxi-out times by airport and time of day**, and avoid pairing with departure slots where historical congestion is high.
- Use this analysis to:
  - Prioritize routes at **airports with shorter taxi-out histories**
  - Avoid **late afternoon slots** at high-traffic hubs where taxi time spikes
  - Add **extra buffers in turnaround planning** for airports with consistently long taxi-outs.This is an area IndiGo doesn't fully control — but understanding and routing around it can save hours of downstream disruption.

Estimated OTAR Impact

- Even a **5-minute reduction in average taxi-out time** can improve OTAR on thousands of flights.
- If IndiGo reroutes just **20% of flights** away from 30+ min taxi-out slots, it could see a **3–4% lift in OTAR**, especially on trunk routes.

For a new entrant in the U.S. market, that's a strong operational differentiator.

Conclusion: Long taxi-out = delayed arrival. The runway might be out of IndiGo's hands, but planning around it isn't. Avoid the gridlock, and you land on time.

Overall Analysis & Strategic Recommendations

1. Fridays are a major weak spot

   The data shows that all top 10 high-risk flights are scheduled for Friday evenings. OTAR scores drop sharply in this window. These flights are not only more likely to be delayed — they're also clustered around major hubs like DFW, DAL, and ORD, where late departures are more likely to trigger domino effects.

2. Late pushback kills punctuality

   Flights that push back late from the gate have a strict OTAR of just ~28%. In contrast, on-time pushbacks hit 83% OTAR. That's a 55% swing. This confirms that departure delay is one of the strongest indicators of late arrivals.

3. Certain airports consistently underperform

   A small group of airports — especially DCA, DFW, and DAL — consistently show up in low-OTAR flights, even when other factors (day, time) are average. These airports seem to carry built-in operational risk.

4. Some time slots carry more risk than others

   High-risk flights tend to concentrate around 5 PM to 7 PM, especially on Thursday and Friday. Even good airports struggle during this period. Meanwhile, morning flights (5 AM–10 AM) perform significantly better across the board.

5. Staffing pressure aligns with delay risk

   The highest volume of high-risk flights is concentrated in a few key time slots. For example, Thursday 5 PM had over 2800 flights flagged with OTAR below 50. This suggests that ops teams may be spread too thin during peak-risk windows.

Recommendations for IndiGo

1. Avoid launching new routes in high-risk time slots

   If a new U.S. route is being planned, do not schedule it for Friday evenings. Rescheduling the same flight to a Saturday or Monday morning could lift OTAR by 25–30 %.

2. Use OTAR score to flag risky flights before they take off

   Flights with OTAR_ESTIMATED under 50 should be monitored proactively. This could mean early gate release, priority maintenance, or alternate aircraft prepped in advance.

3. Deploy extra staff in high-risk time slots only

   The data shows clearly which day-hour combinations generate the most trouble. Use this to shift crew scheduling. Focus extra turnaround staff between 5 PM and 7 PM on Thursdays and Fridays — not across the entire day.

4. Rethink connections from low-performing airports
   Avoid scheduling tight connections out of DCA, DAL, and DFW during high-risk hours. Even a small ground delay here can throw off the next leg by an hour or more.
5. Track staffing windows just like track delays
   If Thursday at 5 PM has many high-risk flights, it should be marked as a high-alert time slot.

References:-

- U.S. Bureau of Transportation Statistics – https://www.transtats.bts.gov
- FAA Air Traffic Reports – https://www.faa.gov/air_traffic/by_the_numbers
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